

RadioLens · Longitudinal Liver CT

Four-study image-derived lesion analysis · dual-channel validation added 25 Aug 2026

Study	Liver volume	Liver-lesion volume	Liver-only burden
25 Dec 2025	1706.42 mL	166.82 mL	9.78%
19 Jan 2026	1667.05 mL	170.34 mL	10.22%
26 Apr 2026	1504.11 mL	97.64 mL	6.49%
23 Aug 2026	1467.57 mL	46.56 mL	3.17%

April → August summary

Automated liver-only segmented lesion volume changed from 97.64 mL to 46.56 mL (-52.3%). Burden changed from 6.49% to 3.17%. April and August contrast timing is closely matched. The former L01 nodal classification was withdrawn: L01 is a segment II/III liver mass and is included here. The true portocaval node remains a separate unsegmented specialist-review target. Because the current automatic mask captured only part of the expert-contoured volume, this percentage is an exploratory software trend, not the true clinical response percentage.

Segmentation uncertainty: the primary August pipeline estimates 46.56 mL, while the independent audit estimated 55.54 mL. The radiologist-workstation screenshots total 105.77 cc on the same current CT. This spread is reported explicitly; none of the automatic estimates is treated as a clinical ground-truth volume.

Tracking safeguards

Tracks are assigned using registered spatial overlap/centroid, Couinaud segment, morphology and vessel topology. One-to-one assignment is preferred; ambiguous matches and split/merge events are explicitly flagged. L04 is a split/merge case in April. A missing separate component does not prove lesion resolution.

Independent validation gate

April-to-August liver-mask registration Dice was 90.3%. The dual-channel gate classified 2 tracks as supported, 6 for review, and 1 as not established. The deterministic channel uses registered voxel overlap, centroid distance and anatomy consistency. The AI channel uses independent repeat segmentation on the August iMAR reconstruction. Algorithmic agreement is not clinical ground truth; all findings require radiologist source-image confirmation.

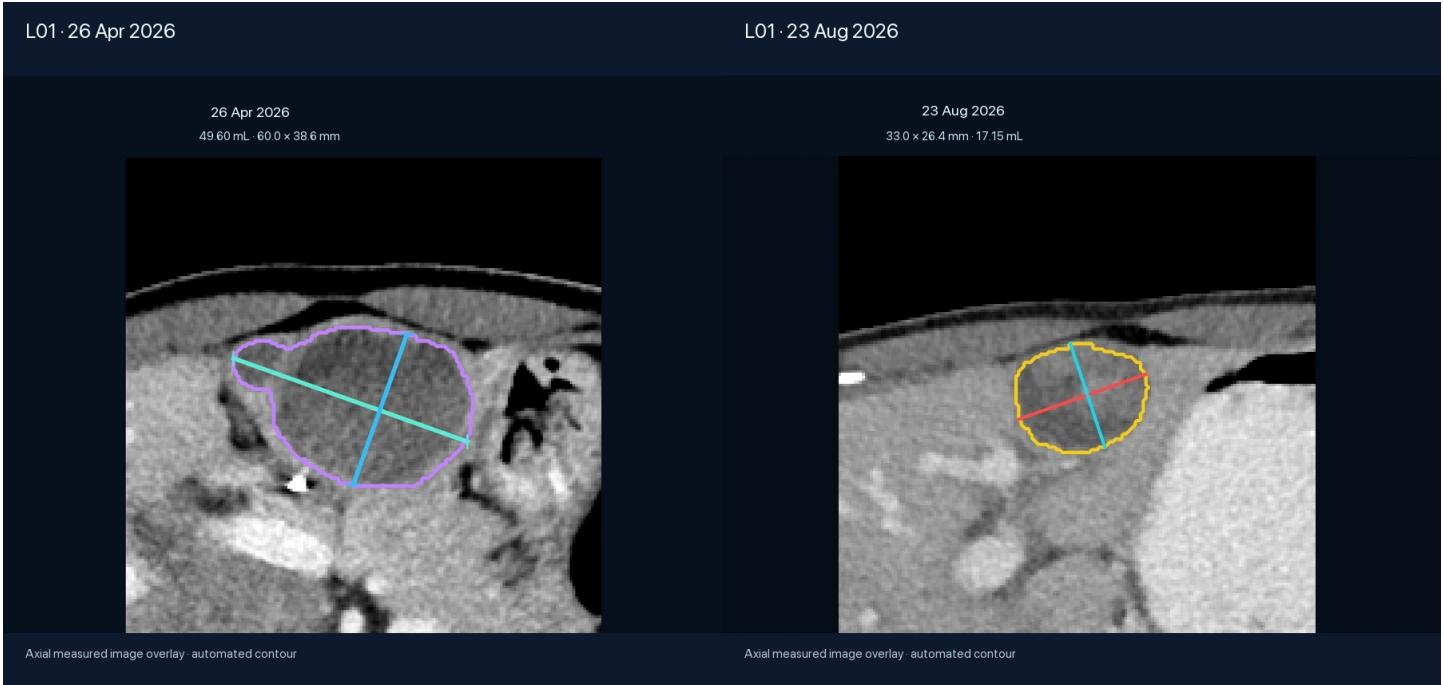
Radiologist-workstation screenshot cross-check

The supplied screenshots display 15 manually segmented targets totaling 105.77 cc on the current 23 Aug 2026 CT, as confirmed by the user. The primary automatic pipeline measured 46.56 mL, only 44.0% of the manual total. If all listed targets are hepatic, their conditional ratio is 7.21% of the automatic liver volume; it is not a final liver-only burden because the manual targets are not anatomically mapped. The current CT screenshot does not contain target names, DICOM coordinates, or an exported segmentation object. Individual T1–T15 contours cannot be safely assigned to longitudinal lesion IDs from screenshots alone.

Target	Volume	Workstation HU display
T1	37.17 cc	77.93 ± 77.93 HU
T2	0.38 cc	75.6 ± 75.6 HU
T3	0.10 cc	50.8 ± 50.8 HU
T4	23.80 cc	72.56 ± 72.56 HU
T5	0.24 cc	79.63 ± 79.63 HU
T6	1.25 cc	89.29 ± 89.29 HU
T7	21.26 cc	55.82 ± 55.82 HU
T8	1.63 cc	70.19 ± 70.19 HU
T9	0.19 cc	63 ± 63 HU
T10	0.07 cc	59.17 ± 59.17 HU
T11	0.93 cc	101.48 ± 101.48 HU
T12	2.34 cc	95.65 ± 95.65 HU
T13	15.33 cc	73.71 ± 73.71 HU
T14	0.21 cc	87.01 ± 87.01 HU
T15	0.87 cc	81.84 ± 81.84 HU
Total	105.77 cc	

Every displayed SD exactly repeats the mean, which is unlikely to be a valid standard deviation. These HU entries are transcribed for audit only and are not used for viability calculations.

L01 · reference segment 3/2

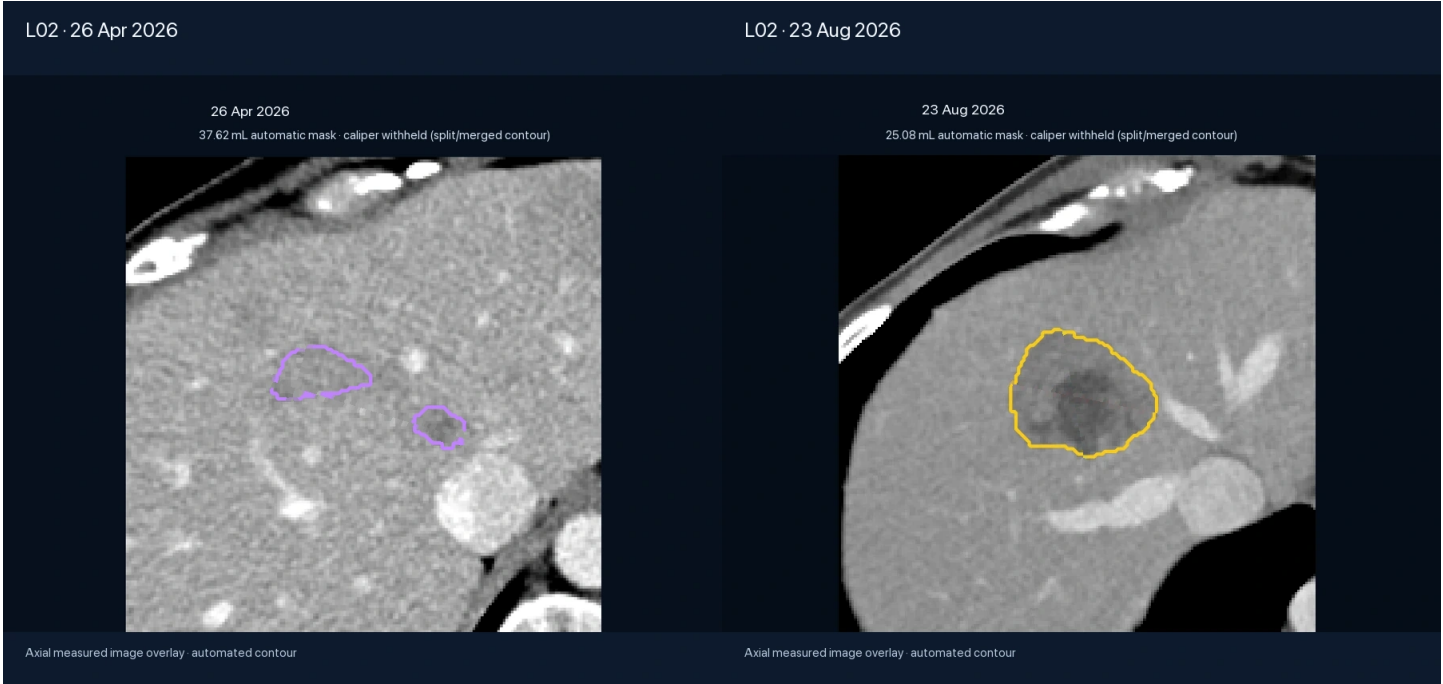


Metric	26 Apr 2026	23 Aug 2026
Detection	Detected	Detected
Volume	49.597 mL	17.154 mL
Long × short × CC	60.0 × 38.6 × 51.0 mm	33.0 × 26.4 × 42.0 mm
VNC-corrected enhancement	10.8 HU	6.2 HU
Below 40 HU	16.6%	33.3%
Validation decision	Supported	Confidence: high
Deterministic geometry	Dice 48.5% · overlap 94.3%	Centroid 7.9 mm · overlapping segment assignment
Independent AI / reconstruction	Pass	Minimum overlap 96.2%

Marked automated decrease in the left-lobe hepatic mass since January. It was previously misclassified as a portocaval node; that label has been withdrawn.

Deterministic: Registered location and mask overlap support this correspondence. AI/reconstruction: The August mask was reproduced on an independently reconstructed CT series.

L02 · reference segment 8

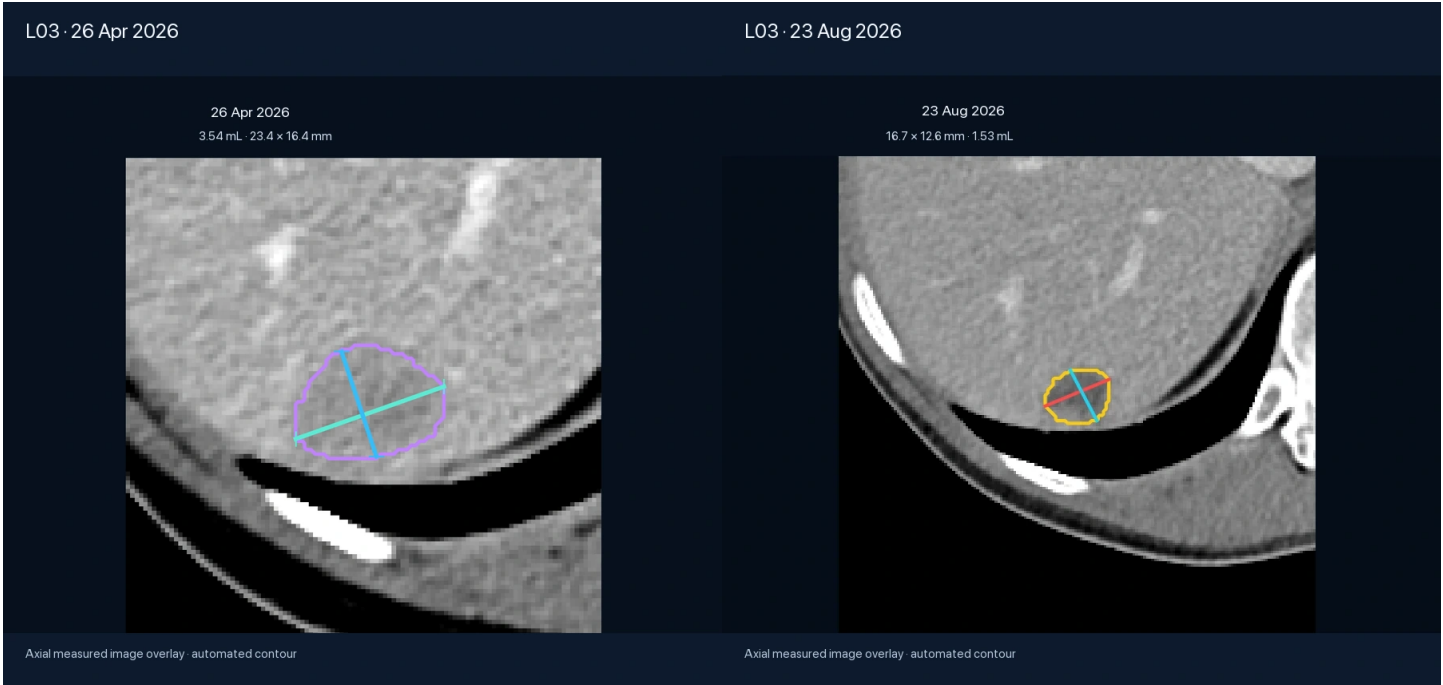


Metric	26 Apr 2026	23 Aug 2026
Detection	Detected	Detected
Volume	37.621 mL	25.080 mL
Long × short × CC	Withheld — split/merged contour	Withheld — split/merged contour
VNC-corrected enhancement	49.5 HU	31.3 HU
Below 40 HU	0.3%	10.1%
Validation decision	Review	Confidence: moderate
Deterministic geometry	Dice 38.8% · overlap 50.0%	Centroid 12.9 mm · reference segment retained
Independent AI / reconstruction	Pass	Minimum overlap 91.4%

Automated volume decrease since January. The residual mask is split into 2 foci.

Deterministic: Main focus is supported; August also contains a reproducible nearby segment 1 focus, treated as a possible split/satellite. AI/reconstruction: Both August components, including the possible satellite focus, were reproduced on iMAR.

L03 · reference segment 7

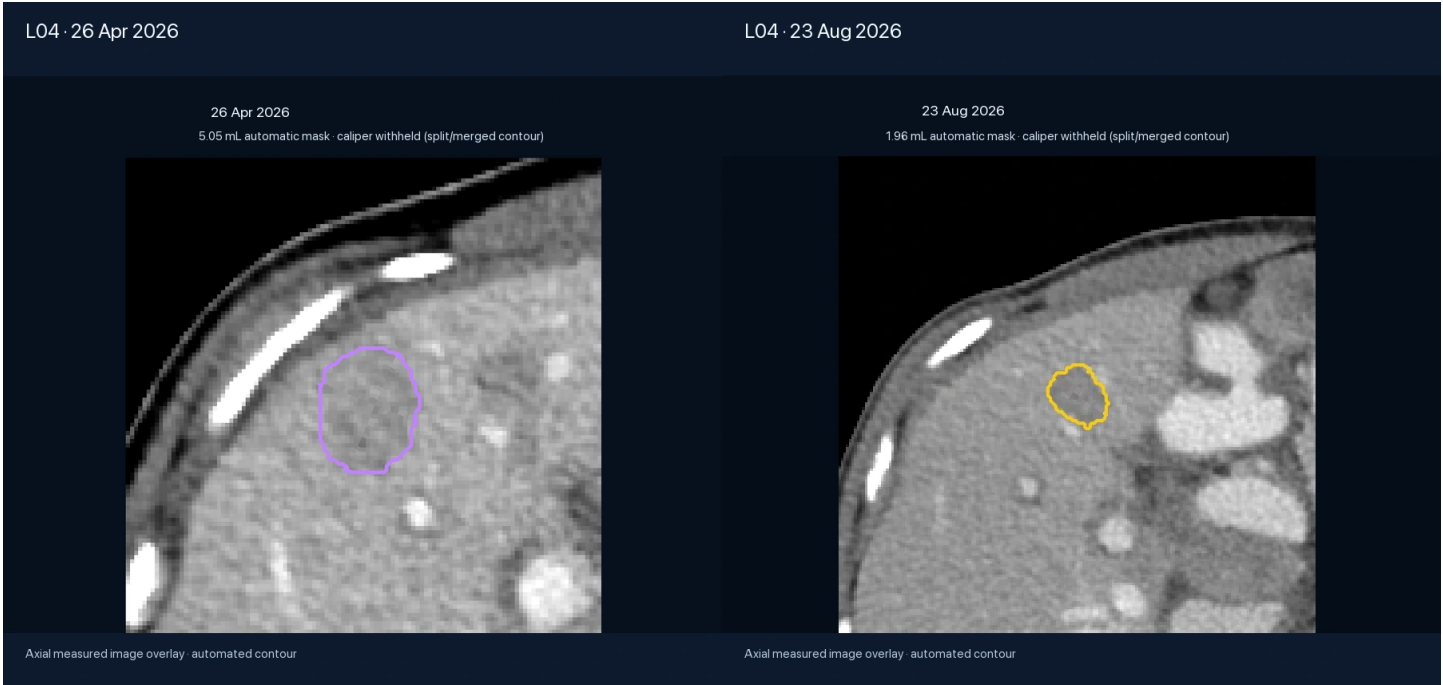


Metric	26 Apr 2026	23 Aug 2026
Detection	Detected	Detected
Volume	3.538 mL	1.534 mL
Long × short × CC	23.4 × 16.4 × 18.0 mm	16.7 × 12.6 × 13.5 mm
VNC-corrected enhancement	57.3 HU	12.0 HU
Below 40 HU	0.0%	18.9%
Validation decision	Supported	Confidence: high
Deterministic geometry	Dice 56.2% · overlap 94.3%	Centroid 2.9 mm · reference segment retained
Independent AI / reconstruction	Pass	Minimum overlap 91.8%

Marked automated volume decrease since January.

Deterministic: Registered location and mask overlap support this correspondence. AI/reconstruction: The August mask was reproduced on an independently reconstructed CT series.

L04 · reference segment 4

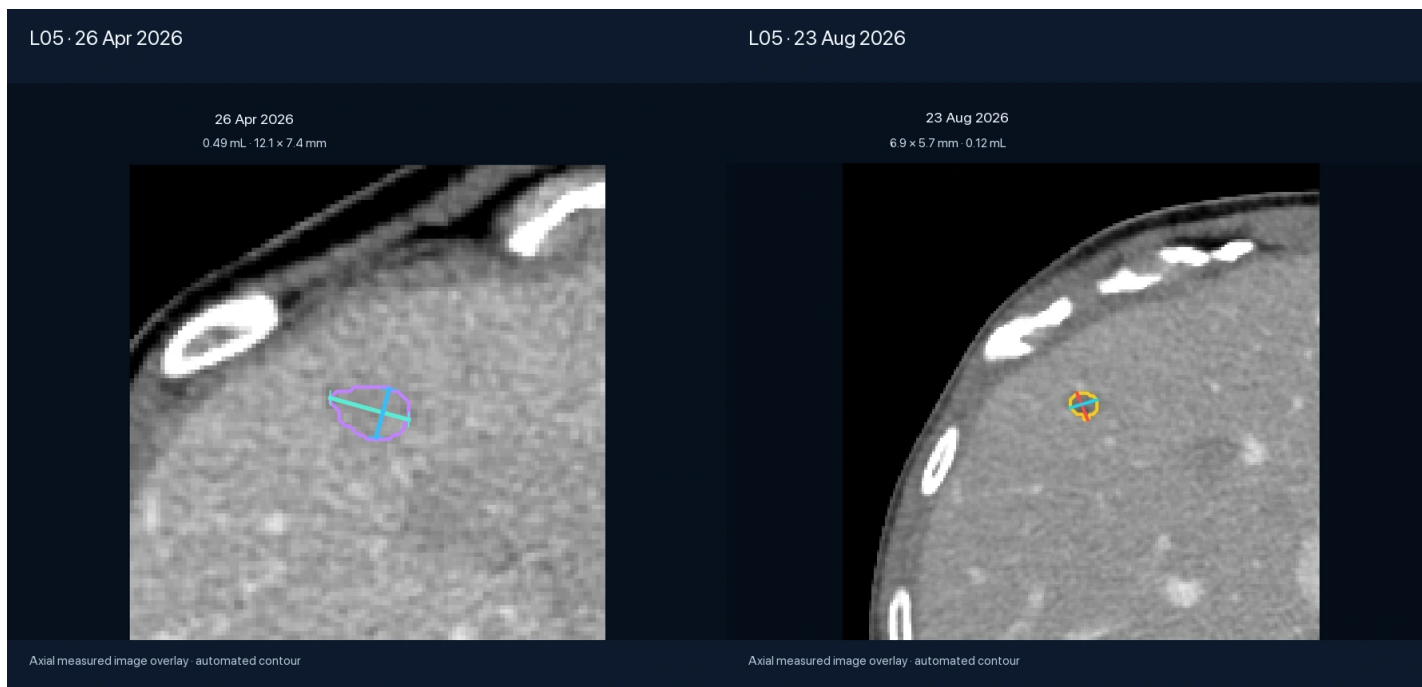


Metric	26 Apr 2026	23 Aug 2026
Detection	Detected	Detected
Volume	5.051 mL	1.956 mL
Long × short × CC	Withheld — split/merged contour	Withheld — split/merged contour
VNC-corrected enhancement	41.6 HU	45.9 HU
Below 40 HU	0.0%	0.0%
Validation decision	Review	Confidence: low
Deterministic geometry	Dice 8.6% · overlap 12.1%	Centroid 9.9 mm · reference segment retained
Independent AI / reconstruction	Review	No independent repeat metric

No large automated volume change. The residual mask is split into 2 foci.

Deterministic: Compound segment 4/8 region with split/merge behavior; do not interpret it as an exact one-to-one lesion measurement. AI/reconstruction: At least one very small series-2 component was not reproduced on the independent iMAR reconstruction.

L05 · reference segment 8

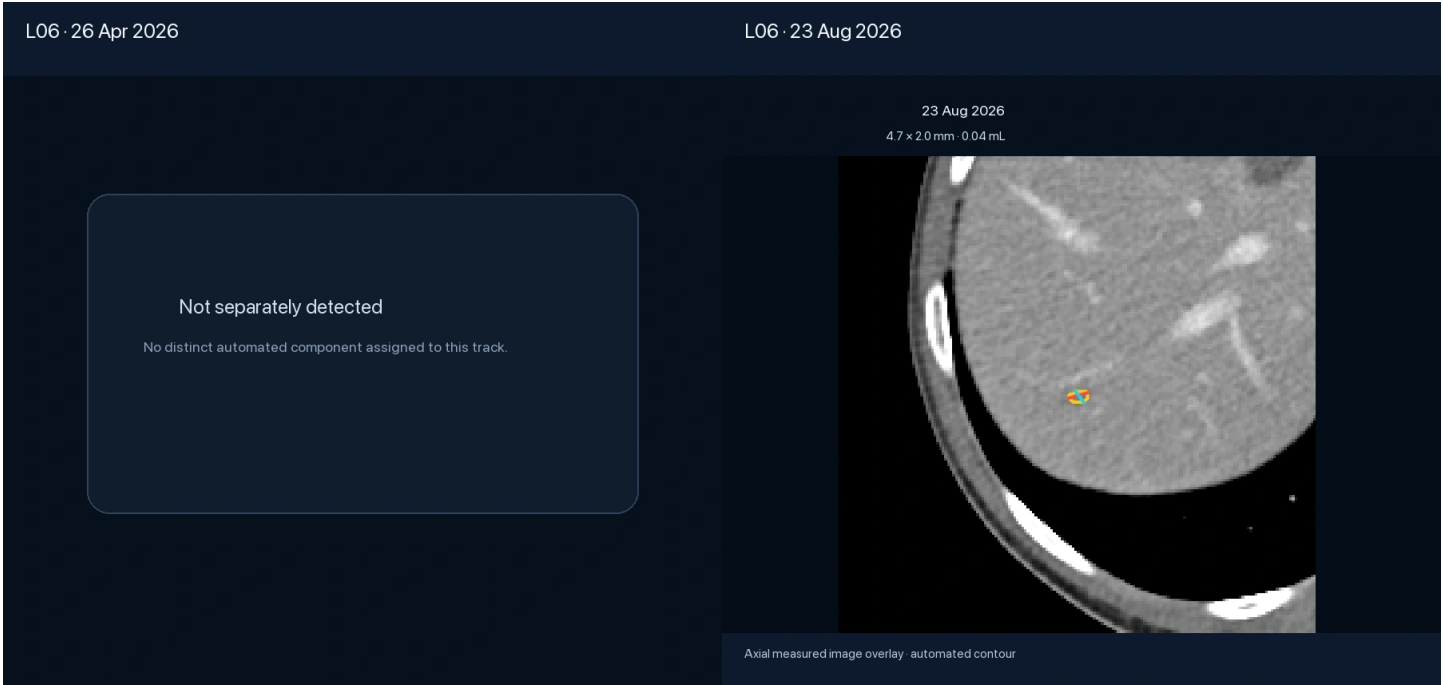


Metric	26 Apr 2026	23 Aug 2026
Detection	Detected	Detected
Volume	0.494 mL	0.118 mL
Long × short × CC	12.1 × 7.4 × 12.0 mm	6.9 × 5.7 × 7.5 mm
VNC-corrected enhancement	47.7 HU	29.7 HU
Below 40 HU	0.0%	5.7%
Validation decision	Review	Confidence: moderate
Deterministic geometry	Dice 0.0% · overlap 0.0%	Centroid 12.3 mm · reference segment retained
Independent AI / reconstruction	Pass	Minimum overlap 83.8%

Marked automated volume decrease since January.

Deterministic: Small focus is anatomically compatible but has no voxel overlap after global liver registration. AI/reconstruction: The August mask was reproduced on an independently reconstructed CT series.

L06 · reference segment 7

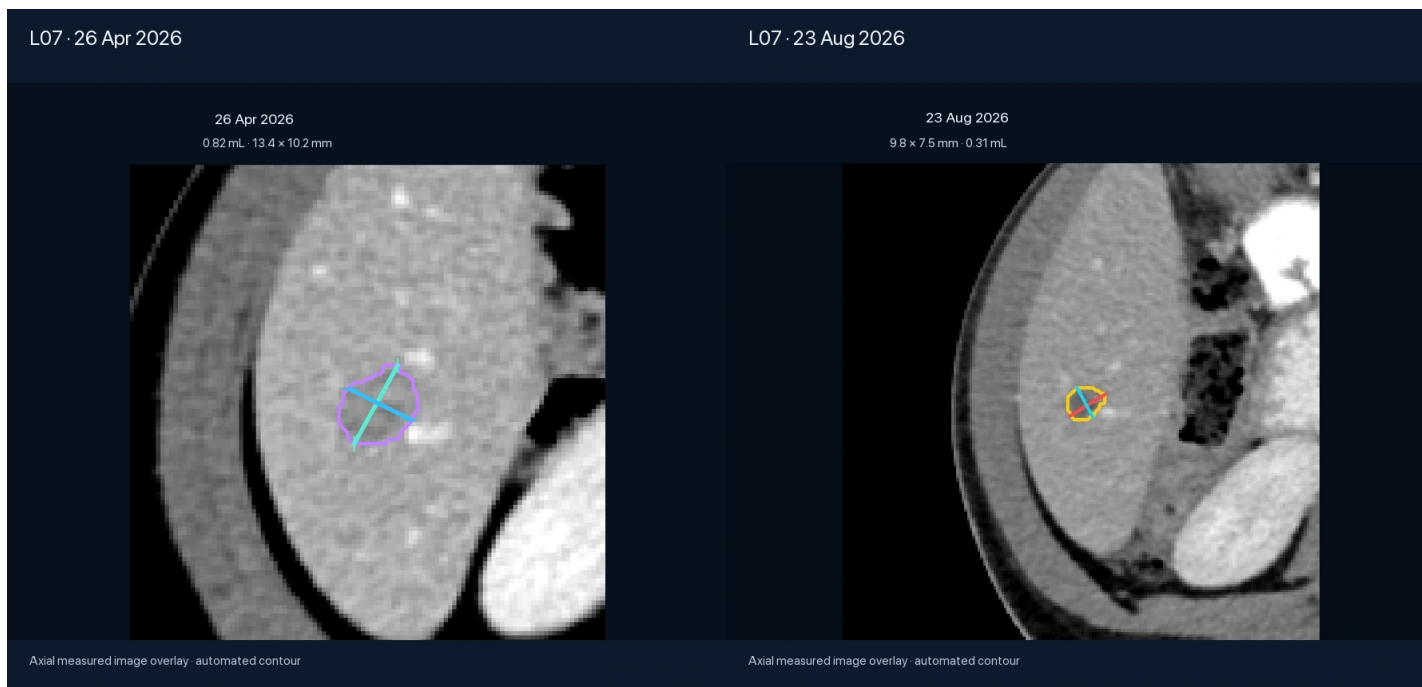


Metric	26 Apr 2026	23 Aug 2026
Detection	Not separate	Detected
Volume	—	0.038 mL
Long × short × CC	—	4.7 × 2.0 × 6.0 mm
VNC-corrected enhancement	—	32.3 HU
Below 40 HU	—	0.0%
Validation decision	Not Established	Confidence: low
Deterministic geometry	No one-to-one match established	not assessable
Independent AI / reconstruction	Review	No independent repeat metric

Marked automated volume decrease since January.

Deterministic: No sufficiently supported one-to-one correspondence was found; absence of a separate component does not prove resolution. AI/reconstruction: At least one very small series-2 component was not reproduced on the independent iMAR reconstruction.

L07 · reference segment 6

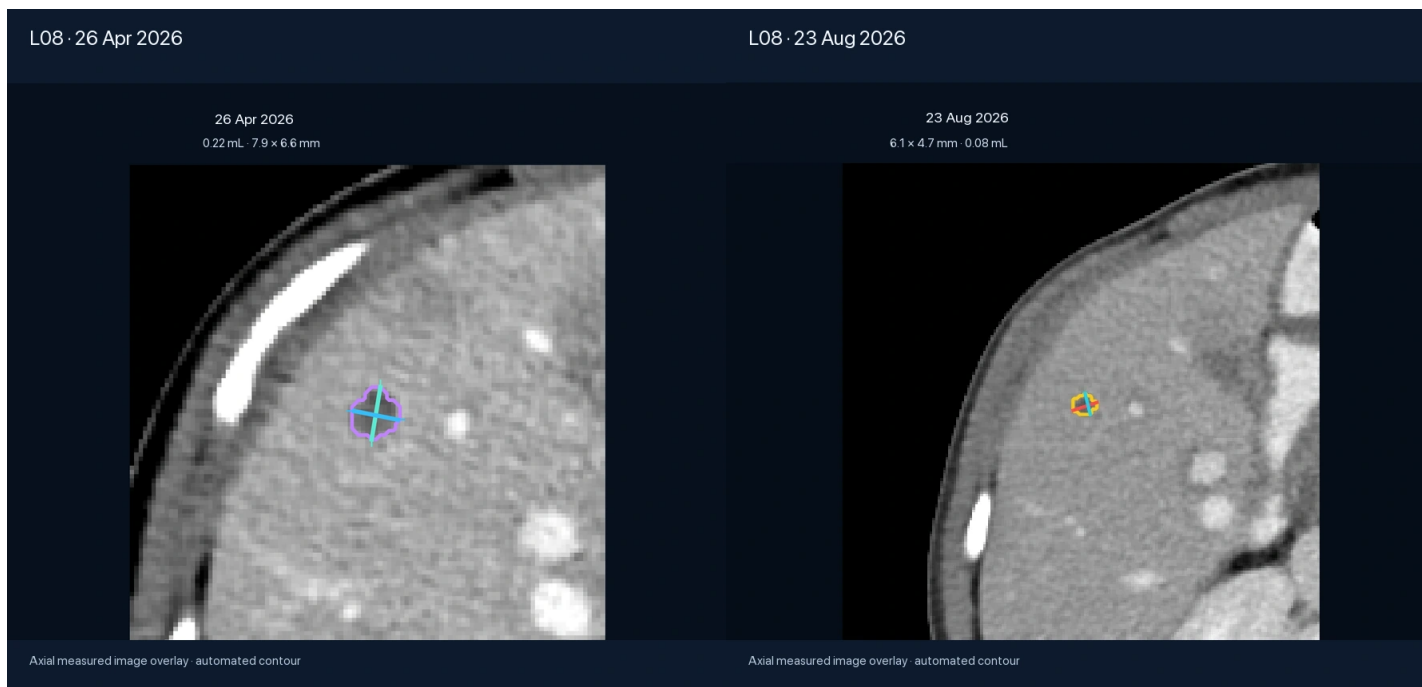


Metric	26 Apr 2026	23 Aug 2026
Detection	Detected	Detected
Volume	0.816 mL	0.311 mL
Long × short × CC	13.4 × 10.2 × 12.0 mm	9.8 × 7.5 × 9.0 mm
VNC-corrected enhancement	49.4 HU	13.1 HU
Below 40 HU	0.0%	17.6%
Validation decision	Review	Confidence: moderate
Deterministic geometry	Dice 0.0% · overlap 0.0%	Centroid 9.9 mm · reference segment retained
Independent AI / reconstruction	Pass	Minimum overlap 89.3%

Marked automated volume decrease since January.

Deterministic: Small focus is anatomically compatible but has no voxel overlap after global liver registration. AI/reconstruction: The August mask was reproduced on an independently reconstructed CT series.

L08 · reference segment 5

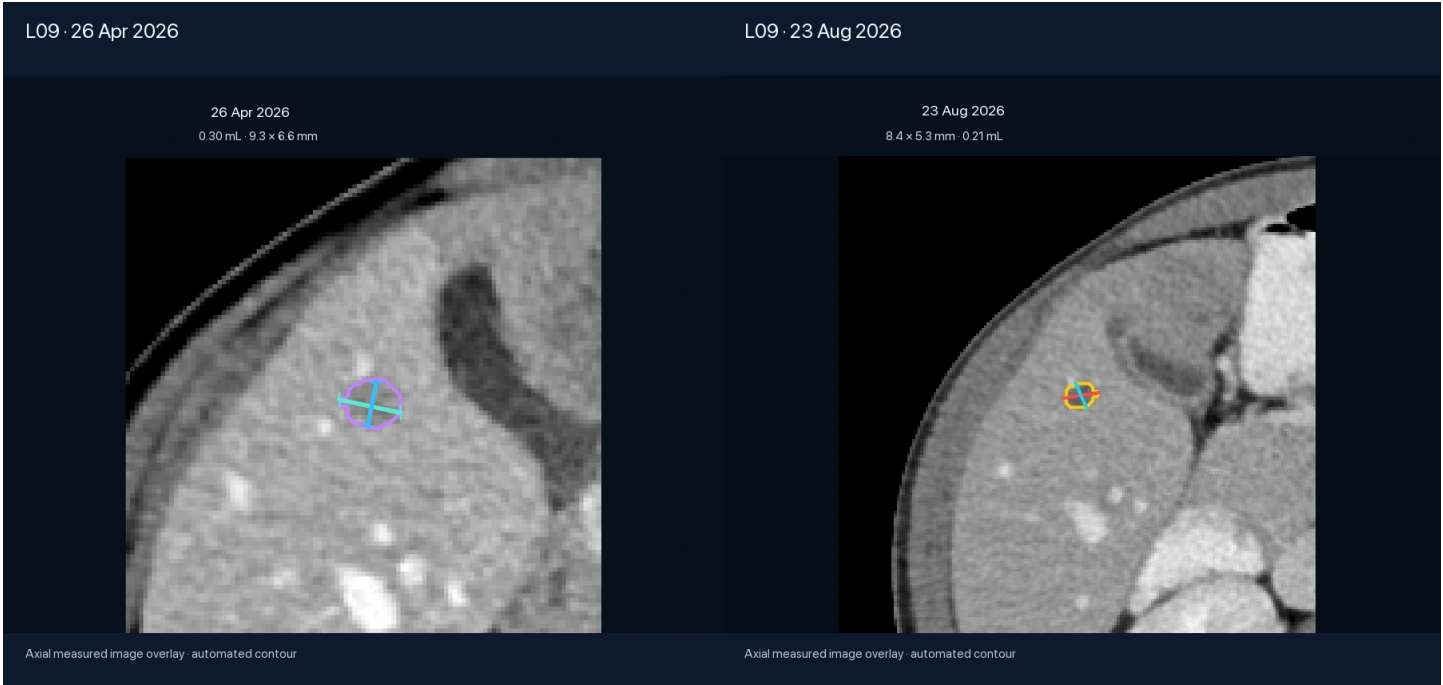


Metric	26 Apr 2026	23 Aug 2026
Detection	Detected	Detected
Volume	0.216 mL	0.077 mL
Long × short × CC	7.9 × 6.6 × 9.0 mm	6.1 × 4.7 × 6.0 mm
VNC-corrected enhancement	6.4 HU	11.2 HU
Below 40 HU	13.3%	8.8%
Validation decision	Review	Confidence: low
Deterministic geometry	Dice 1.7% · overlap 3.0%	Centroid 5.6 mm · reference segment retained
Independent AI / reconstruction	Review	Minimum overlap 78.0%

Marked automated volume decrease since January.

Deterministic: Very small focus; identity is highly sensitive to contour and reconstruction. AI/reconstruction: The August mask was reproduced on an independently reconstructed CT series.

L09 · reference segment 5



Metric	26 Apr 2026	23 Aug 2026
Detection	Detected	Detected
Volume	0.305 mL	0.210 mL
Long × short × CC	9.3 × 6.6 × 9.0 mm	8.4 × 5.3 × 9.0 mm
VNC-corrected enhancement	31.7 HU	-5.6 HU
Below 40 HU	0.0%	25.8%
Validation decision	Review	Confidence: moderate
Deterministic geometry	Dice 32.4% · overlap 41.1%	Centroid 3.3 mm · reference segment retained
Independent AI / reconstruction	Pass	Minimum overlap 85.1%

Marked automated volume decrease since January.

Deterministic: Registered location and mask overlap support this correspondence. AI/reconstruction: The August mask was reproduced on an independently reconstructed CT series.

L10 · reference segment 7



Metric	26 Apr 2026	23 Aug 2026
Detection	Not separate	Not separate
Volume	—	—
Long × short × CC	—	—
VNC-corrected enhancement	—	—
Below 40 HU	—	—
Validation decision	Not Audited	Confidence: unavailable
Deterministic geometry	No one-to-one match established	not assessable
Independent AI / reconstruction	Unavailable	No independent repeat metric

Not separately detected by the automated August model; this does not prove complete resolution.

Deterministic: Not audited AI/reconstruction: Not audited

L11 · reference segment 6



Metric	26 Apr 2026	23 Aug 2026
Detection	Not separate	Not separate
Volume	—	—
Long × short × CC	—	—
VNC-corrected enhancement	—	—
Below 40 HU	—	—
Validation decision	Not Audited	Confidence: unavailable
Deterministic geometry	No one-to-one match established	not assessable
Independent AI / reconstruction	Unavailable	No independent repeat metric

Not separately detected by the automated August model; this does not prove complete resolution.

Deterministic: Not audited AI/reconstruction: Not audited

L12 · reference segment 8



Metric	26 Apr 2026	23 Aug 2026
Detection	Not separate	Not separate
Volume	—	—
Long × short × CC	—	—
VNC-corrected enhancement	—	—
Below 40 HU	—	—
Validation decision	Not Audited	Confidence: unavailable
Deterministic geometry	No one-to-one match established	not assessable
Independent AI / reconstruction	Unavailable	No independent repeat metric

Not separately detected by the automated August model; this does not prove complete resolution.

Deterministic: Not audited AI/reconstruction: Not audited

L13 · reference segment 4/5



Metric	26 Apr 2026	23 Aug 2026
Detection	Not separate	Not separate
Volume	—	—
Long × short × CC	—	—
VNC-corrected enhancement	—	—
Below 40 HU	—	—
Validation decision	Not Audited	Confidence: unavailable
Deterministic geometry	No one-to-one match established	not assessable
Independent AI / reconstruction	Unavailable	No independent repeat metric

Not separately detected by the automated August model; this does not prove complete resolution.

Deterministic: Not audited AI/reconstruction: Not audited

L14 · reference segment 6



Metric	26 Apr 2026	23 Aug 2026
Detection	Not separate	Not separate
Volume	—	—
Long × short × CC	—	—
VNC-corrected enhancement	—	—
Below 40 HU	—	—
Validation decision	Not Audited	Confidence: unavailable
Deterministic geometry	No one-to-one match established	not assessable
Independent AI / reconstruction	Unavailable	No independent repeat metric

Not separately detected by the automated August model; this does not prove complete resolution.

Deterministic: Not audited AI/reconstruction: Not audited

L15 · reference segment 2



Metric	26 Apr 2026	23 Aug 2026
Detection	Not separate	Not separate
Volume	—	—
Long × short × CC	—	—
VNC-corrected enhancement	—	—
Below 40 HU	—	—
Validation decision	Not Audited	Confidence: unavailable
Deterministic geometry	No one-to-one match established	not assessable
Independent AI / reconstruction	Unavailable	No independent repeat metric

Not separately detected by the automated August model; this does not prove complete resolution.

Deterministic: Not audited AI/reconstruction: Not audited

L16 · reference segment 7



Metric	26 Apr 2026	23 Aug 2026
Detection	Not separate	Not separate
Volume	—	—
Long × short × CC	—	—
VNC-corrected enhancement	—	—
Below 40 HU	—	—
Validation decision	Not Audited	Confidence: unavailable
Deterministic geometry	No one-to-one match established	not assessable
Independent AI / reconstruction	Unavailable	No independent repeat metric

Not separately detected by the automated August model; this does not prove complete resolution.

Deterministic: Not audited AI/reconstruction: Not audited